3. Human Systems

Introduction

It is important that the industrial hygiene engineer have a basic understanding of the structure and functioning of the human system. With such an understanding, the industrial hygiene engineer can communicate with other individuals on the health team concerning the probable cause and effect of potential hazard exposures within the industrial environment. Results obtained from the physical examination of workers can provide useful information to the industrial hygiene engineer concerning potential hazards to which the worker may have been exposed. The malfunctioning of the human system may indicate the potential effect of an outside influence, such as the occupational environment.

The occupational environment can affect the performance of the human system in a number of ways. First in the effect to the human system is the physical health of the worker. Many hazard exposures will result in a change in the physical functioning of body systems for those individuals exposed. These changes can be identified by certain symptoms that the worker exhibits. The occupational environment may also affect the emotional health of the worker. This effect is not always as evident as a change in physical health. It is also very difficult to document changes in emotional health and to relate these changes to occupational exposures. The most obvious effect on the human system, and the one in which a cause-and-effect relationship is easiest to identify, is the traumatic injury to the worker. If a worker puts his hand in a band saw and loses three fingers, it is obvious that the effect, the loss of three fingers, was caused by an occupationally related incident. However, it is not always easy to determine the factors that caused the worker to put his hand in the band saw in the first place.

To assess the extent of a hazard, it is necessary to determine the effect of the hazard on the human system. Generally, this effect results in a change in the normal function of a particular system within the human body. Such changes may or may not indicate the existence of a problem. Also, the changes may or may not be occupationally related. However, it is important that problems identified during normal physical examination of workers be investigated to determine if there is a cause-effect relationship between the problem and the occupational environment.

From the above it can be implied that determination of the existence of a problem in the functioning of the human system is the responsibility of the physician. The industrial hygiene engineer should not jump to conclusions concerning symptoms observed in the worker without first referring the worker

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to a qualified physician to determine if these symptoms do indicate a malfunction of the worker's body system. Having determined that a malfunction exists, the industrial hygiene engineer should consult literature prepared by various health researchers concerning the probable cause-and-effect relationship to determine whether the worker's problem can be related to the occupational work environment. Once a probable cause-effect relationship has been established, it is the responsibility of the industrial hygiene engineer to determine if an occupational exposure present within the work environment caused the identified health problem.

In this crucial position, the industrial hygiene engineer must be able to interpret the results obtained by the physician and the health researcher and apply these results to determine if the potential cause is present. When interpreting these results, it is important that the industrial hygiene engineer have a basic understanding of the functioning of the human system and the terminology used to describe the system.

The Study of Human Systems

There are four major classifications of study related to the human system. The first classification is the study of the anatomy. This study involves the form of living systems. The study of anatomy is concerned with the identification of the various components of the body. It is also concerned with the structure and location of these components within the body. Figure 1.3.1 presents some common prefixes and suffixes used in medical terminology related to the human system. Figure 1.3.2 presents some common anatomical terms. Knowledge of these terms will help the industrial hygiene engineer to interpret the common medical terminology that may be encountered.

The second major classification related to the study of the human system is physiology. Physiology involves the study of the function of living systems. In simple terms, physiology is involved with the study of what the system does. It is involved with why the system does what it does and how the system performs its functions. It is important that an understanding of anatomy be obtained to relate the system function with its location, structure, and the terminology used to describe its components.

The study of biochemistry is the third major classification of the study of human systems. Biochemistry is related to the chemistry of living systems. The human body, as well as that of other living organisms, is a chemical system. The body manufactures chemicals and uses these chemicals to perform its various functions. Chemical changes within the body often are indicative of a problem within the human system.

The fourth major category of human system study is biophysics. This discipline is related to the study of physical methods used within human systems. The human system is composed of bones, joints, and muscles. The biophysicist is interested in the workings of these components to accomplish the requirements that are placed upon the body.

Figure 1.3.1

Terminology--the human system.

Common Prefixes			
a- (or an-)	absence of	hypo-	below, deficient
ambi-	both	hystero-	uterus
angio-	tube or blood vessel	in-, intra-	in, inside
ante-	before	leuko-	white
anti-	before	macro-	large
anthro-	joint	mal-	disordered, bad
bi-	two	mamma-	breast
brady-	slow	meno-	monthly, menstrual
cardio-	heart	micro-	small
cephalo-	head	myo-	muscle
cerebro-	brain	nephro-	kidney
chole-	biled	neuro-	nerve
circum-	around	oto-	ear
contra-	against	para-	side
cyt-	sac	pneumo-	air, lung
derma-	skin	poly-	many
dys-	disordered, painful, difficult	post-	after
		pre-	before
		pseudo-	false
en-	in	pulmo-	lung
entero-	intestine	pyelo-	kidney
erythro-	red	pyo-	pus
gastro-	stomach	retro-	behind
glyco-	sugar	rhino-	nose
hem-, hema-, hemato-	blood	semi-	half
		sub-	under
hemi-	half	super-, supra-	above, greater
hepa-, hepato-	liver	tachy-	fast
hydro-	water	trans-	across
hyper-	above, excess		

Common Suffixes			
-algia	pain	-oma	tumor
-asthenia	weakness	-ostomy	opening
-cyte	cell	-paresis	weakness
-ectomy	surgical removal	-pathy	disease
-emia	blood	-phobia	fear
-esthesia	feeling	-pnea	breathing
-genic	causing	-scopy	see
-graphy	visualization	-uria	urine
-itis	inflammation		

Figure 1.3.2

Anatomical terms.

Relating to Location	
anterior (ventral)	toward the front of the body
posterior (dorsal)	toward the back of the body
superior	upper
inferior	lower
superficial	near the surface
deep	remote from the surface
internal	inside
external	outside
proximal	part nearest (with reference to the heart)
distal	part furthest (with reference to the heart)
medial	toward the center of the body
lateral	away from the center of the body; to the side
Relating to Direction	
craniad (cephalad)	toward the head
caudad	toward the feet
Relating to Position and Movement	
supine	lying horizontal on the back, face upward
prone	lying horizontal, face down
abduction	a movement away from the body
adduction	a movement toward the body
flexion	the act of bending, or the condition of being bent
extension	the movement that brings the parts of a limb toward a straight condition

The Basic Unit of Life--The Cell

As the atom is the building block of all substances, so the cell is the building block of all living organisms. All tissue in the human body is made up of individual cells, each of which acts as a site for conversion of nutrients into energy and waste materials. The principal functions of cells are essentially the same regardless of where the cells are located in the human body. These functions are:

A. Exchange of materials with the immediate environment. The exchange

between the environment and the cell occurs by diffusion through the cell membranes.

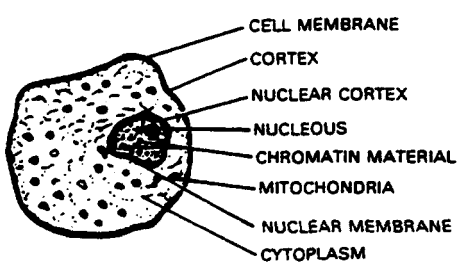
B. The production of energy from nutrients (metabolism). Carbohydrates, fats, and protein are broken down, yielding energy, heat, and a variety of chemical waste products. This breakdown of carbohydrates, fats, and protein is accomplished through the use of oxygen supplied by the blood. The resultant metabolism yields waste products, including carbon dioxide, lactic acid, ketones, and urea.

C. The synthesis of proteins. Within the cell, proteins are built from the various chemicals that are present. These proteins are the building blocks of the body.

D. Reproduction. Reproduction occurs through the division of cells, with some cells reproducing more readily than others. This is evident where regenerative tissue, such as skin, is compared to less regenerative tissue, such as the heart muscle.

Figure 1.3.3

The cell.



All cells, regardless of their location and purpose as a part of the body tissue in which they are present, perform the above functions in the same manner. Specialization of cells occurs when groups are formed together to accomplish different functions of the human body. The general term used for the specialization of these cell groups is tissue. The four main types of tissues in the human body are:

A. Epithelial tissue--the covering of the outer surface of the body and its organs.

B. Connective tissue--the supporting and connecting tissue of the body, including tendons, ligaments, cartilage, and bones.

C. Muscle tissue--the tissue that functions to move the body and its organs.

D. Nervous tissue--the tissue that acts to send messages to and from the various organs and muscles of the body.

The cell is composed of many types of molecules. The most important of these is water which comprises approximately eighty percent of the total weight of the cell. Water is the basis for practically all processes that occur within the cell and thus within the human body. The major constituents of the cell structure are the organic molecules (compounds of carbon) that occur in a large number within the cell. These organic molecules include proteins, lipids (fats), carbohydrates, and the two important nucleic acids, deoxyribonucleic acid (DNA) and ribonucleic acid (RNA). The DNA is the blueprint of genetic information in the cell nucleus, and the RNA transmits this information in forms that can be used by the cell to reproduce itself and its products in the same form. The third class of chemical cell components is inorganic electrolytes such as phosphorus, potassium, sulphur, chlorine, iron and other trace elements.

The basic process which transforms food to energy is as follows. Proteins are broken down into simple amino acids in the digestive system. Carbohydrates and fats are similarly broken down within the digestive system. These chemicals are then absorbed into the blood stream and transported to the various cells of the body. When the nutrients reach the cell, they penetrate the cell membrane through the process of diffusion. Within the cell proper, a process called metabolism occurs. In metabolism, a chemical reaction within the cell fragments large molecules into smaller ones and synthesizes large molecules of a different structure from the raw materials that have been fragmented. During this action, energy is released.

While the cells and tissues within the body are performing their functions, the body remains in system balance, or homeostasis. The rate of metabolism within the body must constantly change in reaction to external stimuli from the environment in which the body is living. The changes that occur within the body to allow the metabolic rate to change must balance each other to assure that the system does not get out of control It is because of homeostasis that, for example, an individual shivers when placed in a cold environment. Shivering is the body's attempt to generate heat through muscle activity to compensate for the lower environmental temperature. Another example of the maintenance of system balance occurs when the human is exposed to an extremely hot environment. The body attempts to maintain a temperature equilibrium by evaporating perspiration and dilating surface blood vessels to allow more blood to pass near the surface of the skin and become cooled. The system balance is very delicate. Although the balance can be maintained within a wide range of certain environmental stimuli, once an imbalance occurs, the result can quickly become a significant danger to the individual.

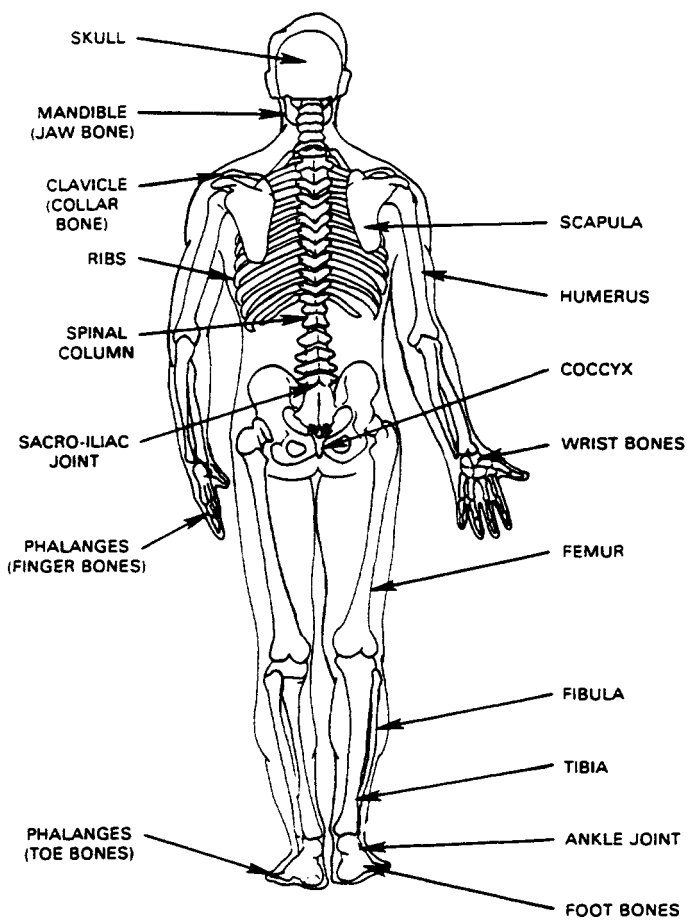
The Structure of the Body--The Skeleton

The structure upon which the human body is built is the skeleton. The skeleton is composed of 206 bones. It provides the basis for the shape of the body as well as protection within its cavities for vital organs, such as the brain, heart, and liver.

The axial skeleton is made up of the bones of the skull, vertebrae, and thorax. The skull contains 29 bones, many of which are fused and appear as one bone structure. The spinal column is made up of 26 bones called vertebrae. The vertebrae serve as the support that enables the body to remain erect. The vertebrae also have a hollow cavity through which the spinal cord passes. Attached to the vertebrae are the 12 pairs of ribs that meet in the center at the breastbone (sternum).

Figure 1.3.4

The skeleton.



The appendicular skeleton consists of the bones of the upper extremities and the lower extremities. The upper extremities include four bones within the pectoral girdle and 60 bones in the arms and hands. The lower extremities consist of two bones within the pelvic girdle and 60 bones within the legs and feet. The bones of the appendicular skeleton provide the structure for the most mobile parts of the body--the arms and legs.

Where bones come together, a joint or articulation is formed. Such articulations occur at the knee and elbow. The joints permit various types of movement. The degree of movement between joints can be very slight--as that which occurs between bones in the skull, or very large--as that which occurs at the thumb. Articulations are classified as follows:

A. Gliding--as in the joints between vertebrae.
B. Hinge--as at the elbow.
C. Ball and socket--occurs in the hip and shoulder joints.
D. Pivot--allows for rotation of the forearm.
E. Ellipsoidal--allows for movements at the wrist.
F. Saddle--allows for the unique movement of the thumb.

There are many types of motions that occur at the joints. A standard nomenclature has been developed for these motions. The following presents the various movements that have been described:

A. Flexion-extension. Flexion is the movement of a joint in which the angle between the bones is decreased, such as bending the arm at the elbow. Extension is the opposite of flexion where the movement increases the angle between the bones, such as straightening the arm.

B. Abduction-adduction. Abduction is the movement of a part away from the center plane of the body or a part of the body, such as lifting the arm outward away from the body. Adduction is the opposite of abduction: a movement toward the center plane of the body or of a part of the body.

C. Supination-pronation. Supination is the turning of the hand so that the palm faces upward, while pronation is the turning of the hand so that the palm faces downward.

D. Rotation. Rotation is a movement in which a part turns on its longitudinal axis, such as turning the head or turning an arm or leg outward or inward.

E. Circumduction. Circumduction involves rotary movements which circumscribe an arc, such as swinging the arm in a circle.

F. Inversion-eversion. Inversion is the movement of the ankle joint in which the sole of the foot is turned inward. Eversion is the movement of the ankle in which the sole of the foot is turned outward.

G. Elevation-depression. Elevation is a movement in which a part is raised, while depression is a movement in which a part is lowered.

For example, the movement of the jaw up and down illustrates elevation-depression.

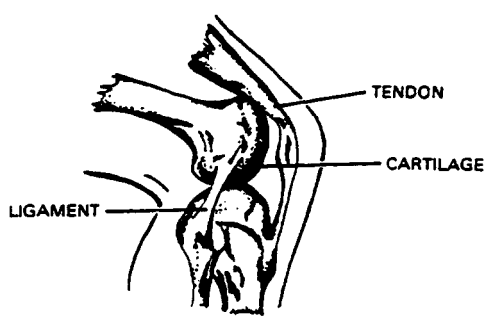
H. Protraction-retraction. Protraction is the movement of a part forward, while retraction is the backward movement of a part. An example is jutting the jaw forward or pulling it backward.

I. Hyperextension. Hyperextension includes movements of the wrist and other joints in which a part is extended beyond the straight line formed by normal extension. An example is a ballet dancer standing on his or her toes.

In addition to the bones of the skeleton, the body is comprised of connective tissue that holds the skeleton together. Three types of connective tissue are present within the body. Tendons, which are cords of strong, elastic fibrous tissue, attach the muscles to the bones. Ligaments, which are bands of tough, flexible connective tissue, join the bones at the articulations. Ligaments tend to inhibit movement beyond a certain point at the joints. Thus, when a joint moves beyond its normal pattern, ligaments may become torn. The third type of connective tissue, cartilage, is a relatively hard, smooth-surface tissue. Cartilage serves as a compression cushion that minimizes jolts and bumps to the skeletal system and protects the bones from breaking or chipping.

Figure 1.3.5

Connective tissue.



The bones of the skeleton have one other function beyond that of providing a structure for the human system. In the center of the bone is a cavity that contains red bone marrow. This bone marrow is important in the production, maintenance, and disposal of blood cells in the adult. Thus, the bone is a living group of cells just like any other part of the human body.

The Moving Force--The Muscles

The muscles are made up of many separate fibers that are able to contract and relax. There are many hundreds of muscles within the human body. If a

muscle is viewed under a microscope, it will appear either as a group of fibers or as an essentially smooth tissue. Those muscles that appear as a group of fibrous tissues are called striated muscles. The skeletal muscles are striated muscles. The smooth-appearing muscles are called smooth muscles. The muscles moving the internal organs within the body are smooth muscles.

There are three major classifications of muscles. The first classification is the voluntary muscle: for example, the biceps, triceps, flexors, and extensors. The muscles are called voluntary because they are set in motion by the conscious direction of the human. A second group of muscles involves the involuntary muscles: for example, the diaphragm, esophageal, and stomach muscles. These muscles act independently of conscious will and thus are called involuntary. The third muscle classification is cardiac muscle. This is the heart muscle. The heart muscle acts as an involuntary muscle, but is generally classified separately from the other involuntary muscles because of its structure: striated muscle joined together in a continuous manner.

Voluntary muscles perform two major functions. The first of these is the maintenance of posture. The voluntary muscles tend to hold the body in the position in which it rests. Certain muscle contraction is required to maintain a given position, which is why a person can get tired while being seated or standing in one position. The second major function of the voluntary muscles is the movement of the human body. By relaxing and contracting, the muscles cause the articulations to move, thus propelling the body or part of the body in the desired direction.

The involuntary muscles provide propulsion of substances through the body passages, as in the case of the movement of blood through the circulatory system and the movement of food from the mouth to the stomach through the esophagus. These muscles also provide for the expulsion of stored substances, such as the action of the intestinal muscles. Other involuntary muscles involve the regulation of opening and sizes of tubes, thus providing for the change in pupil size and the expansion and contraction of blood vessels.

The muscles, whether voluntary or involuntary, receive a stimulus from the brain through the nerves to the muscle. This stimulus causes the muscle to either contract or relax. In order to accomplish this contraction or relaxation, glucose is converted to energy (metabolism). Oxygen is required to burn the glucose in this conversion process. The conversion process results in movement of the muscle that allows work to be performed while much of the energy created is dissipated as heat.

When a muscle performs its function for a long period of time, it is subject to muscle fatigue. Fatigue occurs because of either a lack of nutrients or a lack of oxygen being supplied to the muscle. Thus, the individual operating at a less than adequate nutritional level will feel fatigue much more quickly than an individual who has maintained the proper diet. Lack of oxygen can, in the short run, be compensated for by the body's buildup of an oxygen debt. For this reason, the human, after performing strenuous work, often becomes winded and must breathe hard for a period of time to repay the oxygen debt.

The Control System--The Nervous System

The human body is constantly bombarded by stimuli from the environment. The nervous system acts as the control mechanism that interprets the stimuli received and translates this interpretation into the appropriate action required by the body.

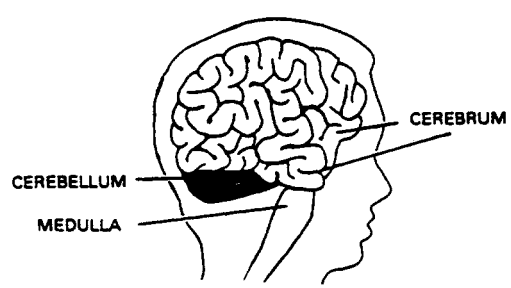
Anatomically, the nervous system can be broken down into two components--the central nervous system and the peripheral nervous system. Functionally, the nervous system can be broken down into the automatic nervous system and the voluntary nervous system.

The central nervous system is composed of the brain and spinal cord. The brain acts as the computer that directs all actions of the human body. The brain has several functions:

A. As a regulatory center. In response to impulses that are received about the state of the body's system and the external environment, the brain sends out messages to enable the system to make the appropriate adjustments.

Figure 1.3.6

The brain.



B. As the seat of sensations. It is the brain that makes us aware of ourselves and our surroundings.

C. As the center of sensations. The individual's perception of the environment received from the various sensory organs is interpreted by the brain. It is the brain that decides that an individual prefers the taste of a medium-rare sirloin steak to that of a hot dog.

D. As a source of voluntary acts. The brain sends signals to the various muscles of the body, causing them to relax or contract depending on the wishes of the individual. These voluntary acts can become programmed, as in typing, so that they appear to be automatic.

E. As a seat of emotions. The brain is the source of emotions, such as happiness, sadness, fear, and all other feelings that affect human behavior.

F. As a seat of higher mental processes. The brain acts as the basis for all thought, reasoning, judgment, memory, and learning. It is through learning that the voluntary act of typing becomes similar to reflex action.

The spinal cord serves as a conducting pathway from the brain to the various parts of the body. It is also a reflex center where actions, such as the well-known knee-jerk reflex, originate.

The peripheral nervous system connects the central nervous system with the distant parts of the body. It acts as a telegraph wire transmitting messages to and from the central nervous system. The sensory nerves or afferent nerves carry impulses from receptors located in all parts of the body to the brain. The motor nerves or efferent nerves carry impulses from the central nervous system to the muscles or other organs that must react to stimuli.

The autonomic nervous system, which is part of the peripheral nervous system, is important in regulating the automatic or involuntary functions of the body. The autonomic nervous system can be divided into two separate systems--the parasympathetic nervous system, which controls the body functions such as heart rate and respiration, and the sympathetic nervous system, which controls the body's response to stress and danger. The parasympathetic nervous system regulates the body functions, while the sympathetic nervous system regulates the body balance or homeostasis.

The voluntary nervous system is the system that carries messages from the brain to the voluntary muscles. Conscious control over the voluntary muscles of the body is maintained by messages transmitted over the voluntary nervous system.

The nervous system works on a combined electrical-chemical process. Although this process is not completely understood, it appears that an electrical impulse is transmitted along the nerve fibers while chemicals help control the rate of the impulses at the junctions of the nerves (synapses).

Fuel Processing--The Digestive System

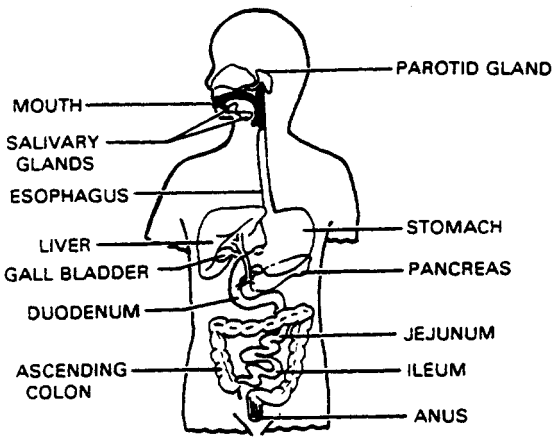
The digestive system acts both chemically and mechanically to transform food into a form that the cells can use. The digestive system is made up of the mouth, including the salivary glands, esophagus, stomach, intestines, liver, gall bladder, and pancreas.

Digestion begins in the mouth. Food is mechanically broken down into small pieces, and the salivary glands release an enzyme that begins the chemical breakdown of starches. Whether our mothers know it or not, they were assisting the digestive process when they told us to chew our food well.

After leaving the mouth, the food travels through the pharynx or throat into the esophagus. The esophagus is the muscular tube that extends from the pharynx to the stomach. Liquids travel through the esophagus by gravity, while solid foods are propelled by muscular contractions of the esophagus.

Figure 1.3.7

The digestive system.



In the stomach, digestion of protein is initiated. The food is mixed with gastric juices containing acid and digestive enzymes, and the stomach walls move to churn the food and mix it with the gastric juices.

After a sufficient time, the food enters the upper part of the small intestine, the duodenum, where the partially digested food is subjected to enzymes secreted into the duodenum by the pancreas. Bile, which is produced in the liver and stored by the gall bladder, also enters the duodenum to aid in the digestion of fats.

The liver, in addition to forming bile, also functions in the metabolism of carbohydrates, the formation of red blood cells and clotting agents, and the detoxification of potentially harmful substances. The detoxification function of the liver is the reason it becomes important in the study of long-term exposures to potentially hazardous chemicals. Quite often these exposures cause pathological changes to occur in the liver, such as an increase in size or a modification of function.

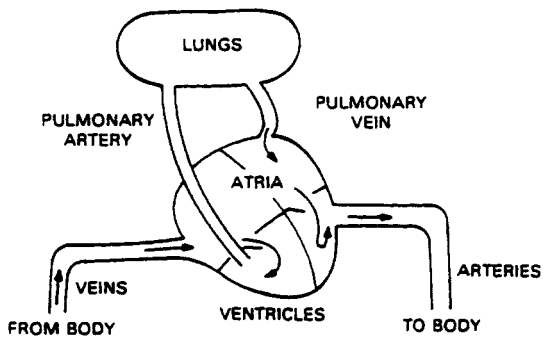
After leaving the upper portion of the small intestine, the foods move through the remainder of the small intestine where the digested proteins, fats, and carbohydrates are absorbed through the intestinal walls into the blood and lymph. The remainder of the waste products pass into the large intestine or colon where water is extracted to produce a semi-solid waste form. This waste is then passed out of the body.

The Distribution System--The Circulatory System

The circulatory system is involved in transporting the body fluids from one region of the body to the other. It is made up of two separate and distinct systems: the blood vascular system and the lymphatic system.

Figure 1.3.8

The circulatory system.



The blood vascular system consists of the heart and associated blood vessels. The heart is a hollow muscular organ that lies between the lungs in the center of the chest directly behind the sternum. The heart is divided into four chambers: two atria and two ventricles. Blood collected from the veins returning from the various portions of the body enters the right atrium. The blood then passes to the right ventricle from which it is pumped through the pulmonary arteries into the lungs. In the lungs, the blood is oxygenated and passes through the pulmonary vein to the heart where it enters the left atrium. From the left atrium, the blood enters the left ventricle where it is pumped through the aorta to be distributed throughout the system.

The circulatory system is composed of veins and arteries. The pulmonary artery previously mentioned carries blood from the heart to the lungs. The pulmonary vein returns the blood to the heart to be distributed through the aorta, arterioles, and capillaries to the body tissues. The venules, veins, and venae cavae return the blood to the heart to begin the cycle once again.

Blood itself is made up of red blood cells, white blood cells, platelets, and plasma. The red blood cells contain hemoglobin that plays the key role in transporting oxygen to the peripheral tissues. The white blood cells are responsible for the body's defense against microorganisms. Platelets are the component of the blood that cause clotting. The remaining liquid of the blood is called plasma.

The second major component of the circulatory system is the lymphatic system. The lymphatic system is made up of capillaries, vessels, and ducts. It also includes special function tissues, such as the lymph nodes, the spleen, the tonsils, and the thymus. The lymphatic system differs from the circulatory system in that it is a one-way system carrying fluids from the peripheral tissues back to the blood vascular system.

Fluid that has filtered out of the blood capillaries into the peripheral tissues must be returned to the blood stream to maintain the plasma volume of the system. This fluid (lymph) is returned to the blood stream by the lymphatic system. Certain special function tissues add lymphocytes and other antibodies to the lymph that, upon entering the blood stream, will be used to fight various infections.

The Combustion Fuel Supply System--The Respiratory System

All living cells require a constant supply of oxygen to carry out the metabolic processes necessary for life. The metabolic process generates waste in the form of carbon dioxide that must be removed from the system. The respiratory system is responsible for providing the oxygen supply and eliminating carbon dioxide waste.

Air enters the respiratory system through the nose and mouth. Near the middle of the nasal cavity, a series of bones (turbinates) is present. The turbinates are covered by a mucous membrane surface and serve to condition the incoming air by adding heat and moisture and trapping bacteria and dust. The turbinates are also coated with cilia or hair-like filaments that wave back and forth to help the mucous membrane to clean the incoming air. The nasal cavity has several small openings that lead to eight sinuses. The sinuses function to equalize the air pressure in the nasal cavity.

The conditioned air that has entered through the nasal cavity passes into the pharynx and into the trachea (windpipe). The opening to the trachea is blocked by the epiglottis during swallowing to prevent food from entering the respiratory system. The trachea divides into the right and left bronchi that further divide and subdivide many times before entering the bronchiole that end in tiny air sacs called alveoli. Each alveolus has a thin membrane wall surrounded by a network of blood capillaries.

Oxygen is supplied to the body tissue by the air that is inspired and passed through the respiratory system to the alveoli. The oxygen within the alveoli diffuses across the membranes and attaches to the hemoglobin in the blood. The blood carries hemoglobin to the various cells of the body where it is used to burn glucose for energy. The waste product of this metabolism, carbon dioxide, is released and carried in the blood stream back to the lungs. Carbon dioxide is transported by attaching to the hemoglobin in the same manner as oxygen and also by entering into a solution with the blood itself. The carbon dioxide is released to the environment from the lungs during expiration.

The primary moving force of the lungs is the diaphragm, a large group of muscle fibers and tendons separating the thorax from the abdomen. When the muscles of the diaphragm contract, the chest cavity is increased; and a partial vacuum is created within the chest that causes air to flow into the trachea, bronchi, and alveoli. Expiration occurs when the diaphragm muscles relax, thus squeezing the lungs and forcing the air out. This respiration normally occurs at a rate of approximately twelve to eighteen respirations per minute, depending on the activity of the individual. The carbon dioxide level within the blood triggers the respiration rate. As a high carbon dioxide level is sensed, the respiration rate increases.

The Filtering System--The Renal System

The renal system or urinary system controls the production and elimination of waste fluids from the body. The system consists of the kidney, bladder, and excretory passages. The kidney acts as a filter for removal of toxic and waste substances from the blood stream. It also is important to the maintenance of the salt and water balance and acid-base balance within the human system.

The bladder serves as a storage area for waste liquids. From the bladder the waste material passes out of the body.

The Defense Systems--Skin and Sense Organs

As is evident from the above, the human system is a complicated group of subsystems that function in unison to permit life. This system must remain in a delicate balance called homeostasis in order to survive. Many dangers in the outside environment can act to modify this system balance, in some cases to the point where the system goes out of control, and life ceases to exist.To protect the body from these outside influences, nature has provided a series of defensive mechanisms that act in various ways to insure that life goes on. The major components of this defensive system are the skin, with its various layers, and the sense organs, including the eyes, tongue, ears, and nose. Other organs of the body such as the liver and the kidney also provide protection to the body as a part of their normal function. The detoxification action of the liver and kidney has been previously discussed. The maintenance of the delicate system balance by the autonomic nervous system and the involuntary muscles as a defense against such hazards as heat or cold has also been discussed. In this section, the sense organs and the skin as defensive mechanisms will be discussed.

The eye, which enables man to see, is the first of these defensive systems. To make vision possible, the structures of the eye convert light rays to nerve impulses that are then transmitted directly to the brain. Since the distance between the eye and the brain is so short, the reaction time between sight and action is quick. The eye itself is protected by a system of lachrymal glands and ducts that secrete tears and convey them to the surface of the eye to remove foreign matter before damage to the lens mechanism occurs. The dilation and constriction of the pupil, controlled by the autonomic nervous system and the involuntary muscles of the eye, allow for sight in areas where limited or excessive light is present.

The ears provide a second defense for the body. Although the eyes cannot see what is going on behind a person, the ears can hear noises that are generated from any direction. Everyone is aware of what happens to an individual when a loud sound occurs immediately behind him. The ear is the first receptor of the initial signal that a potential danger exists. Upon hearing such a sound, the individual turns so that the eyes might determine if a danger does exist. In addition to providing for hearing, the ear also has the function of maintaining the body's physical balance. This is accomplished within the semicircular canals in the inner ear. Further discussion on the anatomy and physiology of the ear will be presented in a later chapter.

The sense of smell, which originates in the olfactory mucosa of the nose, is another defensive mechanism. The sense of smell protects the individual from hazards such as fire or the presence of noxious gases.

The taste buds present on the tongue act as a defensive mechanism and provide the pleasure that accompanies eating a piece of chocolate cake. Certain unpleasant tastes, such as that received from spoiled food, help the individual to protect the digestive system from the invasion of unwanted materials.

The skin is one of the more complex defensive systems in the body. In addition to providing the outer covering that holds the tissues together, it protects against the loss of fluid and heat. The skin acts as the principal barrier between man and the hazards that exist within the environment. Through nerve receptors in the skin, communication between man and the environment is maintained. The skin is made up of the epidermis or outer surface, the dermis or inner surface, hair and hair follicles, sebaceous glands for secretion of oil, sweat glands for secretion of sweat, nerve receptors, fat, small muscles, and blood vessels. Through the skin man is able to experience sensations of pain, touch, heat or absence of heat, traction or pinching, and tickling. The reflex action that occurs when someone touches a hot surface is an example of one of the ways the skin acts as a protection to the body. The sebaceous glands secrete oil that acts as a protective layer to the skin and prohibits the entry of potentially hazardous substances into the body.

In today's modern world we tend to overlook the fact that the sense organs provide primary protection to the human body. However, if one can put himself in place of prehistoric man, the protection provided by these sense organs becomes much clearer. The primary purpose of sight in prehistoric man was not to watch television or read the newspaper but to survey his environment and determine if any dangers were lurking therein. In the same manner, the primary purpose of hearing was not to listen to a politician's speech or to enjoy a symphony orchestra but rather to listen for sounds that indicated the presence of danger. Though these sense organs have found other uses today, they are still important in the defense of the human system.

Other Systems--Reproductive and Chemical Control

For completeness, it is important that two other major systems of the body, the reproductive system and the endocrine system, be mentioned. The reproductive system obviously allows man to continue in existence through the reproduction of the human species. The endocrine system provides the chemical control for the human system.

The endocrine system and the reproductive system are linked together, since many of the organs that are important for reproduction are also important in providing for the chemical balance of the human system. The endocrine system is a very complicated system, the operation of which is not totally understood. We do know that this system of glands exerts a profound

influence on man, his activities, and the metabolic processes within the human body. Some of the functions that the endocrine system performs are:

A. Regulation of growth, development, and sexual maturation of the young.
B. Regulation of water excretion and retention.
C. Regulation of electrolyte salt balance.
D. Regulation of lactation for the new mother.
E. Regulation of sexual development, activity, and reproduction.
F. Regulation of body composition and ability to adapt and resist stress.

The major glands of the endocrine system are the pituitary gland, which acts to regulate growth, water balance, and electrolyte salt balance in the blood; the thyroid, which controls metabolic processes and the rate of metabolic processes; the adrenals, which assist the body in coping with stress; and the gonads, or ovaries and testes, which are essential in the development of secondary sex characteristics.

Summary

The industrial hygiene engineer is required to communicate with medical personnel concerning health problems of workers that may be related to an occupational exposure. During such communications, it is necessary for the industrial hygienist to be familiar with the terminology related to the human system as well as to have a basic understanding of the functions performed by the various systems within the human body. With this information, the industrial hygiene engineer has a basis for determining the relationship between the conditions exhibited by the worker and the potential exposure to a particular occupational hazard.